

Def.

Let  $E \subset X$  be a non-empty subset of a Metric space  $X$ . Define

$$P_E: X \rightarrow \mathbb{R}; x \mapsto \inf_{z \in E} d(x, z)$$

is called a distance from  $x$  to  $E$ .

Prop. Let  $P_E$  be a distance from  $x$  to  $E$  in a Metric space  $X$ .

1).  $P_E(x) = 0$  if and only if  $x \in \overline{E}$ .

2).  $P_E$  is Continuous Uniformly.

Proof. To show the first, using the property of closure and infimum

$$x \in \overline{E} \Leftrightarrow \forall \varepsilon > 0, B_\varepsilon(x) \cap E \neq \emptyset \Leftrightarrow \forall \varepsilon > 0, \exists y \in E \text{ such that } d(x, y) < \varepsilon \Leftrightarrow \inf_{z \in E} d(x, z) = 0.$$

To show the second assertion, observe that: given  $x, y \in E$ ,

$$P_E(x) \leq d(x, z) \leq d(x, y) + d(y, z) \text{ for any } z \in E$$

Since it holds for all  $z \in E$ ,

$$P_E(x) \leq d(x, y) + P_E(y)$$

and which implies  $P_E(x) - P_E(y) \leq d(x, y)$ . Similarly,  $P_E(y) - P_E(x) \leq d(x, y)$ , so

$$|P_E(x) - P_E(y)| \leq d(x, y) \text{ for any } x, y \in X.$$

Given  $\varepsilon > 0$ , take  $\delta = \varepsilon$ , then we obtain that

$$d(x, y) < \delta \Rightarrow |P_E(x) - P_E(y)| \leq d(x, y) < \varepsilon$$

□

Lemma. Let  $A, B \subseteq X$  be non-empty disjoint closed set in Metric space  $X$ .

Then, a map  $f: X \rightarrow \mathbb{R}; x \mapsto \frac{P_A(x)}{P_A(x) + P_B(x)}$  where  $P_E: X \rightarrow \mathbb{R}; x \mapsto \inf_{z \in E} d(x, z)$

is continuous function satisfying  $A = f^{-1}[\{0\}]$  and  $B = f^{-1}[\{1\}]$ .

Proof. At first, since

$$P_A(x) + P_B(x) = 0 \Leftrightarrow P_A(x) = P_B(x) = 0 \Leftrightarrow x \in \overline{A} \cap \overline{B} = A \cap B = \emptyset,$$

so  $P_A(x) + P_B(x) \neq 0$  for all  $x \in X$ , so  $f$  is well-defined

And, since  $P_A$  and  $P_B$  are continuous,  $P_A + P_B$  also continuous, so  $f = P_A / (P_A + P_B)$  is continuous, being rational of continuous function.

Observe that: since  $P_A + P_B > 0$  and  $P_A, P_B \geq 0$ ,

$$f(x) = 0 \Leftrightarrow P_A(x) = 0 \Leftrightarrow x \in \overline{A} = A \quad \text{and}$$

$$f(x) = 1 \Leftrightarrow P_B(x) = 0 \Leftrightarrow x \in \overline{B} = B, \quad \text{therefore}$$

$$A = f^{-1}[\{0\}] \quad \text{and} \quad B = f^{-1}[\{1\}], \quad \square$$